

When Experimental-Plan Scores Miss Task Loss: A Diagnostic Study of LLM Bayesian Design

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Abstract

Large language models can propose rich experimental plans, but deploying those plans requires a scorer whose rankings survive contact with realized task loss. We study this link in multi-source location finding with an explicit likelihood. An entropy-based StrategyEIG scorer ranks realized entropy reduction ($\rho = 0.38\text{--}0.44$), yet is nearly uncorrelated with realized localization RMSE and does not beat naive or greedy-EIG execution. We then test the natural repair: replace entropy with expected posterior permutation-matched RMSE. A pre-registered, no-LLM replay over 60 standard-task probe states fails its proceed gate: the depth-1 task-loss scorer reaches macro Spearman $\rho = 0.231$ against realized posterior-risk reduction, below the ≈ 0.3 threshold. Increasing rollout integration to 1024 samples and repairing particle-support coupling do not recover the gate. We therefore stop before policy deployment. The result is a measured negative finding: task-aligned utilities are not sufficient when posterior-predictive plan values weakly rank truth-conditional outcomes.

1 Introduction

LLM-driven Bayesian experimental design (BED) separates an attractive pair of capabilities. A language model can express a reusable strategy in natural language, while an explicit simulator can score the strategy before execution. This separation promises non-myopic planning without training a task-specific policy or expanding a full LLM action tree. It also creates a fragile link: the quantity that is easy to estimate in simulation may not rank the quantity the user ultimately cares about.

Information gain is the standard BED objective. For localization, however, posterior concentration is not the same as localization accuracy. A posterior can lose entropy by concentrating on the wrong branch of a multimodal source configuration. This is not merely a philosophical distinction: in our banked Path A experiments, entropy-scored plans rank realized entropy and truth-log-posterior quality, while their correlation with point RMSE is near zero and the resulting StrategyEIG policy trails naive and greedy EIG.

This paper asks whether replacing entropy with a task-aligned posterior loss repairs that first decision link. We use expected posterior permutation-matched RMSE, which is available at decision time and never accesses the hidden source configuration. Before implementing or deploying an arbitration policy, we pre-register a local ranking-fidelity gate on existing standard-task candidate roots. The gate fails, including after rollout-count, horizon, and support-resampling diagnostics. Following the pre-registration, we do not launch the downstream policy pilot.

Our contributions are:

1. A two-link diagnostic that separates scorer-to-posterior-objective fidelity from posterior-objective-to-task-loss rankability.
2. A measured entropy/task-loss mismatch for LLM-scaffolded experimental design, supported by paired Path A policy results.
3. A no-LLM test of expected posterior RMSE as a repair on standard multi-source location finding, including convergence and support checks.
4. A negative-result protocol in which a failed scorer gate prevents an expensive policy claim rather than being hidden by a downstream sweep.

2 Related Work

Bayesian experimental design. Deep adaptive design amortizes sequential BED policies and evaluates total information gain with contrastive bounds [Foster et al., 2021]. Classical decision-theoretic design also permits utilities defined by downstream loss rather than entropy. Our setting keeps the location likelihood explicit so that the utility question can be isolated from learned-posterior error.

LLM-driven design. BED-LLM uses language models for intelligent information gathering [Choudhury et al., 2025]. Brute-force LLM lookahead can be non-myopic but grows rapidly with depth. Strategy-level rollout instead asks an LLM to propose natural-language policies and uses simulation to rank them. We focus on the validity of that ranking, not on claiming a new decision-theoretic utility.

Online planning. Constrained BED and informative path planning optimize explicit transition models. COPEX is directly adjacent [Guo, 2026]. The constrained Path A environment in this paper is diagnostic background: its purpose is to show that a planning gap can exist while an LLM scaffold still fails to exploit it. The Path B gate itself uses standard unconstrained multi-source location finding and no environment engineering.

3 Diagnostic Framework

3.1 Location Posterior And Task Loss

A hidden configuration θ contains two source locations in $\mathbb{R}^2$. At query x , the signal is an inverse-square sum with multiplicative log-normal noise. A finite particle support $\{\theta_i\}$ is reweighted with the exact likelihood. Source labels are exchangeable, so configurations are canonically ordered before averaging and all RMSEs use best-permutation matching.

For posterior p , let $\hat{\theta}(p)$ be its canonical posterior mean. We define the decision-time posterior risk

$$L(p) = \mathbb{E}_{\theta \sim p} \left[\text{RMSE}_{\text{perm}}(\theta, \hat{\theta}(p)) \right].$$

For a candidate root query x , the estimated utility is the Monte Carlo posterior-predictive reduction

$$\hat{U}(x) = L(p_t) - \frac{1}{R} \sum_{r=1}^R L(p_{t+1}^{(r)}).$$

This utility is task-aligned but truth-free: simulated truths are sampled from p_t . The realized smooth target fixes the recorded hidden truth, generates paired observation noise, updates the same exact-likelihood posterior, and measures the resulting reduction in L . We also report realized point-RMSE reduction to the fixed truth.

3.2 Gate 0 Protocol

We reuse 60 probe records from 20 trials at history lengths 0, 3, and 6. Each record contains the hidden state, realized history, and 6–10 LLM-proposed strategy roots. Legacy records do not contain rollout posterior snapshots or future query trajectories, so we use the pre-specified local fallback: score the stored roots with the analytic location model and no LLM calls. The canonical result uses depth 1, where there is no continuation-policy mismatch.

The posterior approximation begins with a common pool of 100,000 normal-prior particles, conditions on each history, systematically resamples 256 particles, and applies likelihood-targeted MCMC rejuvenation. Candidate comparisons share random controls. The canonical scorer uses 256 posterior-predictive rollouts and the smooth target averages 256 truth-conditional deployments.

For each probe we compute Spearman correlation across candidate roots and then macro-average over probes. The pre-registered proceed threshold is $\rho \gtrsim 0.3$ at a tested horizon. We additionally report (i) the old entropy scorer against the new smooth target where available, (ii) task scorer against point-RMSE

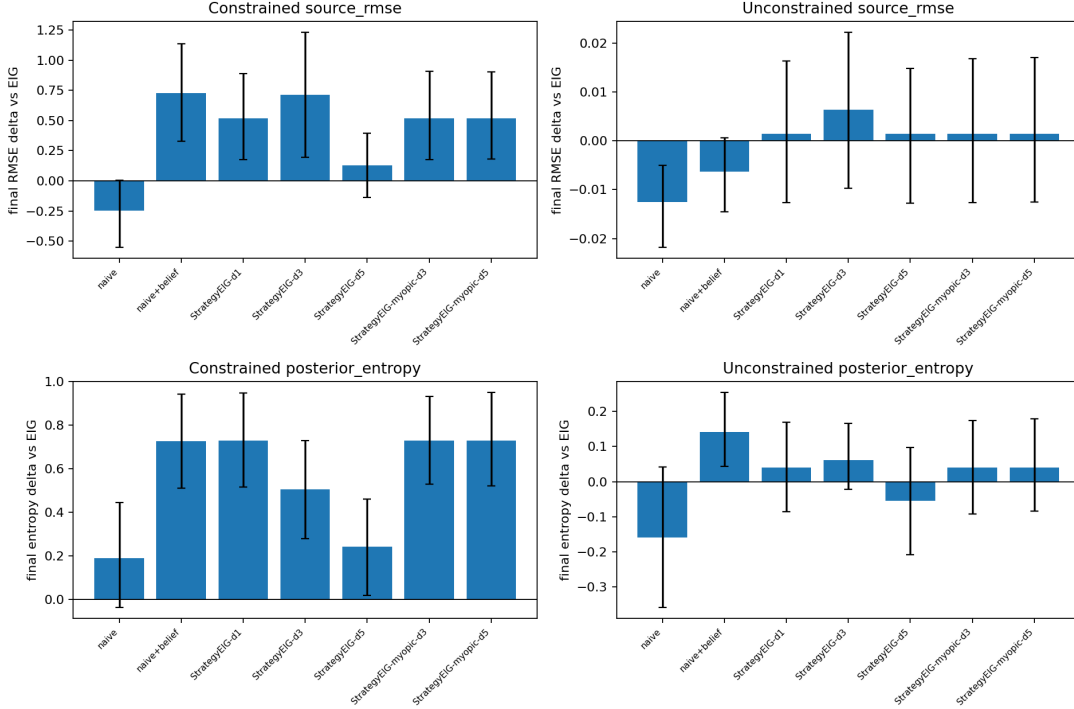


Figure 1: Banked Path A result. Deeper entropy scoring improves the StrategyEIG scaffold in the constrained arm, but StrategyEIG remains worse than naive and greedy EIG. The unconstrained contrast is flat.

reduction, and (iii) the rankability ceiling: realized smooth-target reduction against realized point-RMSE reduction. If the task scorer fails, no arbitration policy or cluster pilot is run.

3.3 Path A Motivation

Path A evaluates entropy-scored StrategyEIG on a locality-constrained branch-decoy environment. It uses common rollout controls, fixed-support analytical posterior updates, support-grid candidate generation, and matched-compute myopic controls. Thirty paired trials compare naive, greedy EIG, and StrategyEIG depths 1, 3, and 5. A second 30-trial arm removes the movement constraint. These runs motivate the repair; they are not evidence for the Path B task-loss method.

4 Results

4.1 Entropy Fidelity Does Not Imply Localization Fidelity

The 26B A4B entropy scorer appears successful under an information-only gate. Across depths 2, 3, and 5, estimated-vs-realized entropy Spearman correlations are 0.380 ± 0.045 , 0.440 ± 0.041 , and 0.396 ± 0.044 ; corresponding truth-log-posterior correlations are about 0.36–0.37. Yet point-RMSE correlations are -0.088 ± 0.051 , -0.080 ± 0.045 , and -0.035 ± 0.050 . The scorer ranks its own information objective but not the localization endpoint.

The paired policy sweep confirms that this disconnect matters. In the constrained arm, final RMSE means are 0.166 for naive, 0.412 for greedy EIG, 0.542 for StrategyEIG-d5, 0.929 for d1, and 1.127 for d3. Depth 5 improves over shorter StrategyEIG and matched-compute myopic controls, but it does not beat either baseline. In the unconstrained arm, all methods are near the task ceiling (0.089–0.108 RMSE) and depth has no meaningful effect.

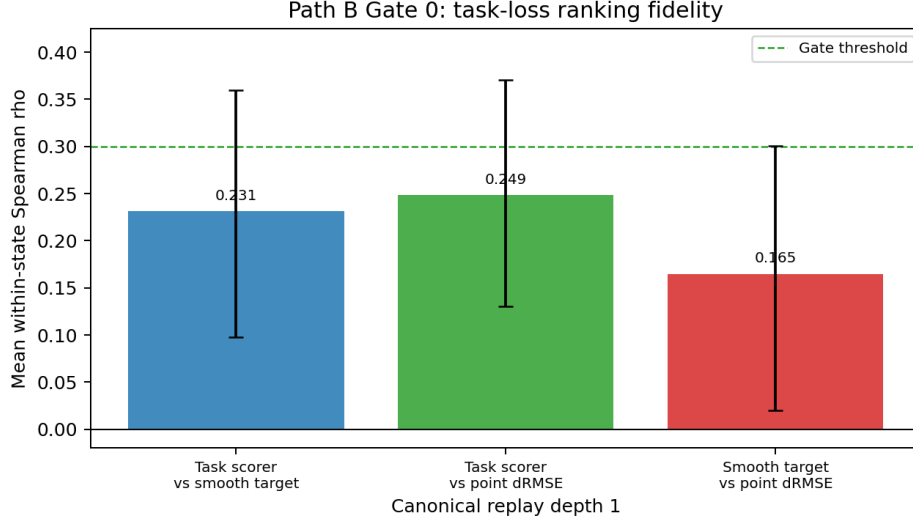


Figure 2: Path B Gate 0. Macro-average within-probe Spearman correlation for the task-loss scorer and its realized targets. The dashed line is the pre-registered proceed threshold. The canonical depth-1 scorer remains below it, so the policy pilot is not launched.

4.2 Task-Loss Repair Fails Its Gate

Figure 2 shows the canonical Path B Gate 0 result. At depth 1, the task-loss scorer reaches $\rho = 0.231$ against realized posterior-risk reduction, with a 95% bootstrap interval $[0.098, 0.360]$. This is positive but below the ≈ 0.3 proceed threshold. Correlation with realized point-RMSE reduction is 0.249 $[0.130, 0.371]$. The rankability ceiling between realized posterior-risk and point-RMSE reductions is only 0.165 $[0.020, 0.301]$.

The failure is not explained by one unlucky integration budget. With independently resampled supports, depth-1 task-loss correlations are 0.278 at 256 samples and 0.262 at 1024 samples. The mean score-SE to between-candidate-SD ratio falls from 0.316 to 0.162 , but fidelity does not improve. A common prior pool and deterministic systematic resampling yield the canonical 0.231 result. Longer horizons also fail: depth 3 reaches 0.145 in the 256-sample diagnostic.

The round split identifies the break. Task-vs-smooth correlation is 0.055 at the prior state, 0.305 after three observations, and 0.335 after six. Once data concentrate the posterior, posterior-predictive utility becomes modestly useful. Before data, however, the scorer averages over possible truths while each realized ranking is conditional on one unknown draw. More rollouts estimate the Bayesian average more precisely; they do not make it predict which prior draw occurred. Selecting only the later rounds would pass, but that stratum was not pre-registered and would omit the first decision faced by the deployed policy.

4.3 Why The Stop Matters

The proposed downstream method would have included the plain raw-history naive query, greedy-EIG query, and LLM strategic roots, then defaulted to naive unless another candidate exceeded it by one rollout standard error. That conservative rule can control estimator noise, but it cannot repair a scorer whose rankings are weakly related to realized gains. Gate 0 therefore prevents a costly and statistically ambiguous policy sweep. No result in this paper claims that task-loss arbitration beats naive or greedy EIG.

5 Discussion

The experiments expose two distinct failures. Entropy StrategyEIG has good scorer-to-objective fidelity but poor objective-to-RMSE alignment. Expected posterior RMSE fixes the semantic objective, yet its posterior-predictive values still weakly rank truth-conditional outcomes, especially before observations. The

second failure is subtler: the utility can be Bayes-rational in expectation without being a reliable per-instance arbiter among candidate LLM plans.

This distinction changes how LLM-BED systems should be evaluated. Reporting only selected EIG or entropy drop can certify the simulator rather than the task. Reporting only final RMSE can obscure whether failure comes from proposal quality, score noise, or an unrankable endpoint. A useful diagnostic chain is: estimated score to realized smooth objective, smooth objective to point task loss, and finally policy improvement under paired deployment. Each link needs its own uncertainty and stop rule.

The negative result does not show that task-aligned utilities are useless. Later-round states cross the nominal correlation threshold, suggesting that arbitration may become viable after posterior concentration. A future method could explicitly abstain at early rounds and activate only after an online fidelity or effective-sample-size criterion. That is a new, post-hoc hypothesis, not a claim tested here.

6 Limitations

This is a workshop-scale diagnostic based on one model family and 60 probe states from a limited number of paired trials. The standard-task Gate 0 uses an exact analytic likelihood, so it isolates utility fidelity under a particle posterior; open-ended LLM-BED settings must also contend with LLM-estimated posterior error. The legacy records omit rollout posterior snapshots, requiring a local fallback. Depth 1 avoids future-policy mismatch, but the common 100,000-particle pool, 256-particle support, and MCMC rejuvenation remain a numerical approximation. Low late-round importance effective sample size is reported rather than treated as exact posterior coverage.

Path A uses one deliberately constructed constrained environment family, method-blind oracle selection, a robustness heatmap, and a limited number of paired trials. Its 26B A4B runs use millions of tokens and include forced-thinking exits. The minimum package lacks an MPC-style direct action-sequence ablation; that remains future work. Most importantly, Gate 1 is not missing positive evidence: it was deliberately not run after the pre-registered Gate 0 failure. Therefore we make no performance claim for the unimplemented arbitration policy and no claim that task-aligned utilities transfer for free to LLM-estimated posteriors.

7 Conclusion

An LLM experimental-design scaffold can rank entropy reduction while failing to rank localization accuracy. Replacing entropy with expected posterior RMSE is principled but does not automatically repair the per-instance selection link: our standard-task scorer remains below its pre-registered fidelity gate after rollout and support diagnostics. The scientifically useful result is the stop itself. Before asking whether deeper LLM plans beat naive or greedy design, we should establish that their scores rank the realized task-facing quantity; in this study, neither entropy nor its task-loss replacement clears that bar robustly.

References

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